

High Gain Observer Demonstration with a Double Pendulum System

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1 Introduction

Modern control systems heavily rely on accurate knowledge of a system's state to function effectively. However, in many real-world cases, it can be extremely difficult to measure every variable needed for control [1]. Often times sensors are too expensive, too bulky, or unavailable for certain physical quantities. State observers are particularly useful in these cases. State observers are defined as mathematical constructs that estimate unmeasured system states from the information that is available [1]. The high-gain observer configuration has gained significant attention for its ability to accurately map system states and disturbances even when dealing with model uncertainty [1]. Through tuning observer gains, the estimation error can be reduced to near zero in finite time domains, making it a critical tool for disturbance rejection.

This project demonstrates the design and implementation of a high-gain observer using a planar double pendulum system. The double pendulum acts as an ideal test platform, as it exhibits nonlinear and chaotic dynamics when acted upon by an external force. A double pendulum is also relatively simple to produce and has a well-established set of motion equations. This system is operated using the SpeedGoat real-time data acquisition and control platform, allowing for smooth data collection of a physical system while evaluating the ability of a calculated control model. The goal of this project is to experimentally evaluate the performance of a high-gain observer system and develop an understanding of translating a predicted model into a functional real-time control system.

2 Computational Analysis

To develop a control model of the system, first the equations of motion (EOM) for a standard double pendulum system were derived using Lagrange's method, with several simplifying assumptions being made. Motion was constrained to a single vertical plane, reducing the system to a two-dimensional problem. The

two rods were also assumed to be of equal length but different masses. This is because the model needs to incorporate the two rods individually so it can model the system both with and without the second pendulum for testing purposes. Small angle approximation also applies, meaning that the displacement of both links were assumed to remain small enough such that $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$. This linearizes the resulting EOM and simplifies analysis while maintaining a reasonable approximation.

2.1 Lagrangian Analysis

The Lagrangian energy method is well suited for multi-body systems like the double pendulum, as it avoids the need to explicitly resolve internal forces [2]. The Lagrangian is defined as

$$\mathcal{L} = T - V \quad (1)$$

where T is the total kinetic energy and V is the total potential energy of the system [2]. Applying the Euler-Lagrange equation to each standard coordinate q_i :

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} \right) - \frac{\partial T}{\partial q_i} + \frac{\partial V}{\partial q_i} + \frac{\partial D}{\partial q_i} = Q_i \quad (2)$$

The generalized coordinates chosen to describe the system were the angular displacements of each rod, θ_1 and θ_2 , measured from the vertical.

Analyzing the system's velocity components and the effect of the external torque, the following total kinetic and potential energy equations were determined to be

$$V = m_1 g \left[\frac{l_1}{2} \cos(\theta_1) \right] + m_2 g [l_1 \cos(\theta_1) + l_2 \cos(\theta_2)] \quad (3)$$

$$T = \frac{1}{2} \left[\frac{1}{3} m_1 (l_1)^2 \right] \dot{\theta}_1^2 + \frac{1}{2} \left[\frac{1}{12} (m_2) (l_2)^2 \right] \dot{\theta}_2^2 + \frac{1}{2} (m_2) (\bar{V}_B)^2 \quad (4)$$

Velocity of the center of mass of the second pendulum is represented as

$$(\bar{V}_B)^2 = (\bar{V}_{Bx})^2 + (\bar{V}_{By})^2 \quad (5)$$

where the position components of the second pendulum's center of mass are:

$$\bar{X}_B = -l_1 \sin(\theta_1) - \frac{l_2}{2} \sin(\theta_2) \quad (6)$$

$$\bar{Y}_B = l_1 \cos(\theta_1) + \frac{l_2}{2} \cos(\theta_2) \quad (7)$$

Taking the time derivatives yields the velocity components:

$$\bar{V}_{Bx} = -l_1 \cos(\theta_1)(\dot{\theta}_1) - \frac{l_2}{2} \sin(\theta_2)(\dot{\theta}_2) \quad (8)$$

$$\bar{V}_{By} = -l_1 \sin(\theta_1)(\dot{\theta}_1) - \frac{l_2}{2} \sin(\theta_2)(\dot{\theta}_2) \quad (9)$$

Substituting these expressions, the squared velocity term becomes:

$$(\bar{V}_B)^2 = \left[-l_1 \cos(\theta_1)(\dot{\theta}_1) - \frac{l_2}{2} \sin(\theta_2)(\dot{\theta}_2) \right]^2 + \left[-l_1 \sin(\theta_1)(\dot{\theta}_1) - \frac{l_2}{2} \sin(\theta_2)(\dot{\theta}_2) \right]^2 \quad (10)$$

Substituting $(\bar{V}_B)^2$ back into the kinetic energy expression gives:

$$T = \frac{1}{2} \left[\frac{4}{12} (m_1)(l_1)^2 + m_2(l_1)^2 \right] (\dot{\theta}_1)^2 + \frac{4}{24} (m_2)(l_2)^2 (\dot{\theta}_2)^2 + \frac{1}{2} (m_2)(l_1)(l_2)(\dot{\theta}_1)(\dot{\theta}_2) \cos(\theta_1 - \theta_2) \quad (11)$$

Linear viscous damping was assumed at the first joint, with the dissipation function taken as $D = \frac{1}{2}c_1\dot{\theta}_1^2 + \frac{1}{2}c_2(\dot{\theta}_2 - \dot{\theta}_1)^2$, where c_1 and c_2 are the damping coefficients. No damping was assumed at the second joint. Applying the Euler-Lagrange equation to each generalized coordinate yields:

For θ_1 :

$$\frac{d}{dt} \left[\frac{\partial T}{\partial \dot{\theta}_1} \right] - \frac{\partial T}{\partial \theta_1} + \frac{\partial V}{\partial \theta_1} + \frac{\partial D}{\partial \dot{\theta}_1} = M \quad (12)$$

For θ_2 :

$$\frac{d}{dt} \left[\frac{\partial T}{\partial \dot{\theta}_2} \right] - \frac{\partial T}{\partial \theta_2} + \frac{\partial V}{\partial \theta_2} + \frac{\partial D}{\partial \dot{\theta}_2} = 0 \quad (13)$$

Evaluating each partial derivative and simplifying, the equation of motion for θ_1 was obtained [2]

$$(m_1 + m_2)l_1^2\ddot{\theta}_1 + m_2l_1l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_2 + m_2l_1l_2\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + (m_1 + m_2)gl_1 \sin(\theta_1) + (c_1 + c_2)\dot{\theta}_1 - c_2\dot{\theta}_2 = \tau \quad (14)$$

Similarly, evaluating the partial derivatives for θ_2 yields the equation of motion for the second coordinate:

$$m_2l_2^2\ddot{\theta}_2 + m_2l_1l_2 \cos(\theta_1 - \theta_2)\ddot{\theta}_1 - m_2l_1l_2\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + m_2gl_2 \sin(\theta_2) - c_2\dot{\theta}_1 + c_2\dot{\theta}_2 = 0 \quad (15)$$

These two equations of motion can be compactly expressed in matrix form as:

$$\mathbf{M}(\theta)\ddot{\theta} + \mathbf{C}(\theta, \dot{\theta})\dot{\theta} + \mathbf{D}\dot{\theta} + \mathbf{g}(\theta) = \tau \quad (16)$$

where the inertia matrix $\mathbf{M}$, Coriolis matrix $\mathbf{C}$, damping matrix $\mathbf{D}$, gravity vector $\mathbf{g}$, and input torque vector $\boldsymbol{\tau}$ are defined as:

$$\mathbf{M} = \begin{bmatrix} (m_1 + m_2)l_1^2 & m_2l_1l_2c_{12} \\ m_2l_1l_2c_{12} & m_2l_2^2 \end{bmatrix} \quad (17)$$

$$\mathbf{C}\dot{\boldsymbol{\theta}} = \begin{bmatrix} m_2 l_1 l_2 s_{12} \dot{\theta}_2^2 \\ -m_2 l_1 l_2 s_{12} \dot{\theta}_1^2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \quad (18)$$

$$\mathbf{g} = \begin{bmatrix} (m_1 + m_2)gl_1s_1 \\ m_2gl_2s_2 \end{bmatrix}, \quad \boldsymbol{\tau} = \begin{bmatrix} \tau \\ 0 \end{bmatrix} \quad (19)$$

where $c_{12} = \cos(\theta_1 - \theta_2)$, $s_{12} = \sin(\theta_1 - \theta_2)$, and $s_i = \sin(\theta_i)$.

2.2 Simulink Model

To validate the derived equations of motion and support the design of the high-gain observer, a simulated model of the double pendulum system was developed in MATLAB Simulink. This model was constructed directly from the matrix form of the EOMs, allowing open-loop pendulum dynamics to be simulated and verified before involving a physical system. This model provided an effective way of confirming that the simulated system exhibited physically consistent behavior. The model also provided a controlled environment in which the observer could be designed and tuned, as gains could be easily adjusted iteratively.

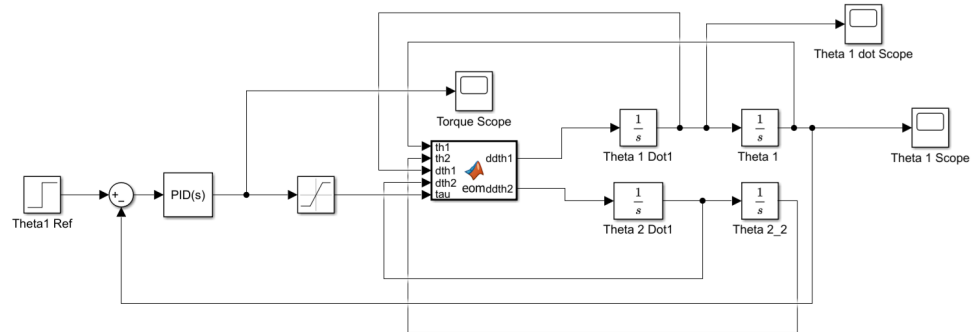


Figure 1: Simulink model of the double pendulum system.

2.3 PID Control and Gain Tuning

To drive θ_1 to a desired reference angle, a PID controller was implemented in Simulink using the parallel form compensator:

$$u(t) = P \cdot e(t) + I \int e(t) dt + D \frac{N}{1 + N \frac{1}{s}} e(t) \quad (20)$$

where $e(t) = \theta_{1,ref} - \theta_1$ is the tracking error and $N = 100$ is the derivative filter coefficient. The controller was implemented in continuous time, and the gains were tuned manually through iterative simulation. The final tuned gains were estimated to be approximately $P = 20$, $I = 5$, and $D = 2$.

At the center of the Simulink model is a MATLAB function block, labeled `eomddth2`, which takes the current states $\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2$ and the applied torque τ as inputs and returns the angular accelerations, $\ddot{\theta}_1$ and $\ddot{\theta}_2$. These are computed by solving for the accelerations:

$$\ddot{\theta} = M(\theta)^{-1} \left[\tau - C(\theta, \dot{\theta})\dot{\theta} - D\dot{\theta} - g(\theta) \right] \quad (21)$$

The outputs $\ddot{\theta}_1$ and $\ddot{\theta}_2$ are then passed through two cascaded integrator blocks to recover $\dot{\theta}_i$ and θ_i for each link, with the states fed back into both the function block and the PID controller. A saturation block was also placed between the PID output and the function block to limit the input torque to a physically realistic range.

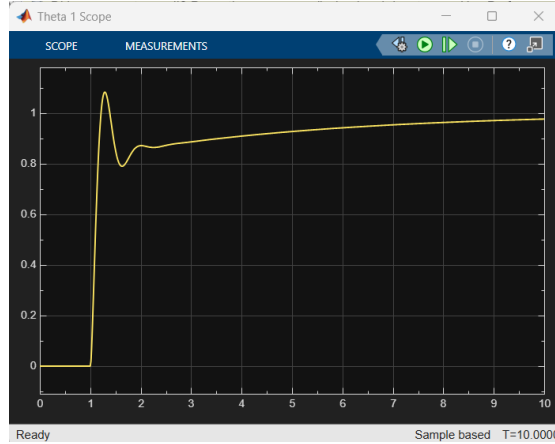


Figure 2: Theta 1 Scope from Simulink Model.

Figure 2 shows the simulated response of θ_1 to a unit step reference input. The system exhibits a small overshoot before settling smoothly to the reference value, demonstrating that the tuned PID gains produce stable and well-damped closed-loop behavior. This confirmed that the equations of motion were correctly implemented and acting as expected.

3 Physical System Development

3.1 Mechanical Design and CAD

The physical double pendulum was designed using computer-aided design (CAD) software prior to fabrication. The two pendulums were designed as straight aluminium bars, 10 inches each. The CAD assembly helped verify the geometry of the system and overall clearances before parts were manufactured. This design was set up to be a tabletop system, so a custom shaft mount and motor mount were also designed to securely interface the brushed DC motor with the pendulum shaft. All of the custom mechanical components were fabricated by the Virginia Tech Aerospace Engineering Machine Shop. A complete drawing of the system model is shown below:

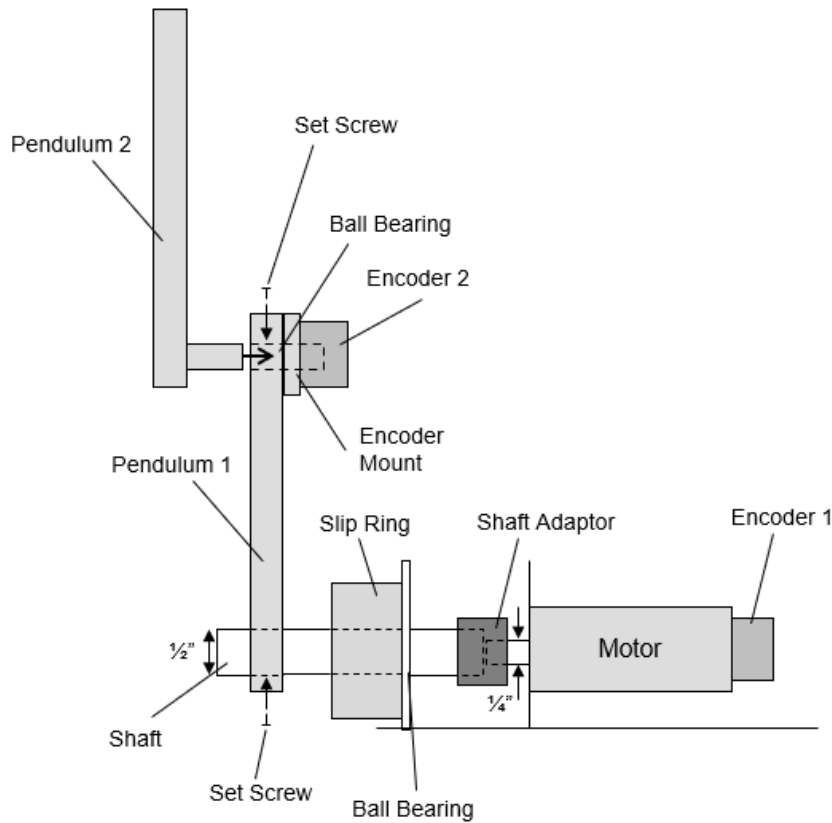


Figure 3: Labeled Assembly Diagram

A slip ring was used to be able to connect the encoders successfully without getting the electronics tangled during operation. As a result, a slip ring mount

was also designed to fully interconnect the system, as seen in the figure below.

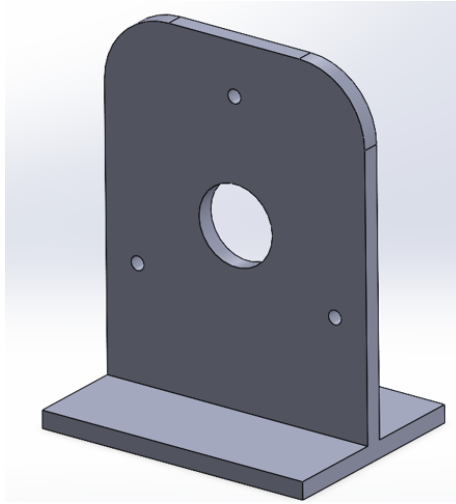


Figure 4: Labeled Assembly Diagram

The pendulums themselves were also designed with CAD as shown in the engineering drawing below, which was submitted to the machine shop for fabrication.

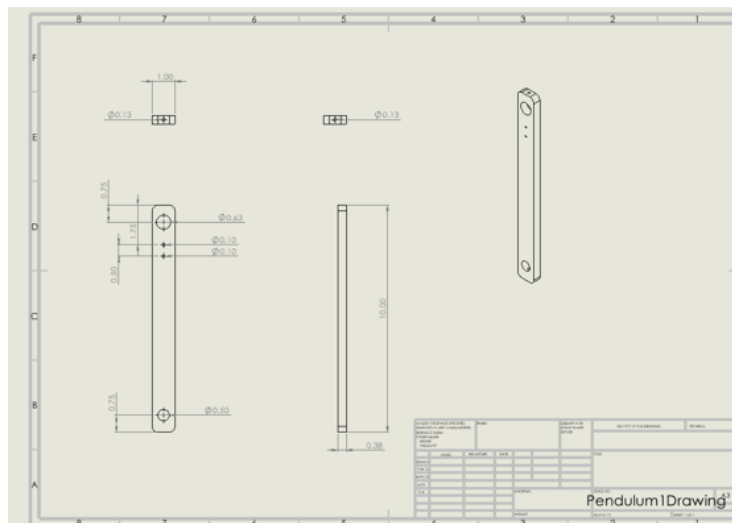


Figure 5: CAD Drawing of the Pendulums

An encoder mount was also designed to sit at the connection of the pendulums and allow the encoder to rotate on the same path as the second pendulum.

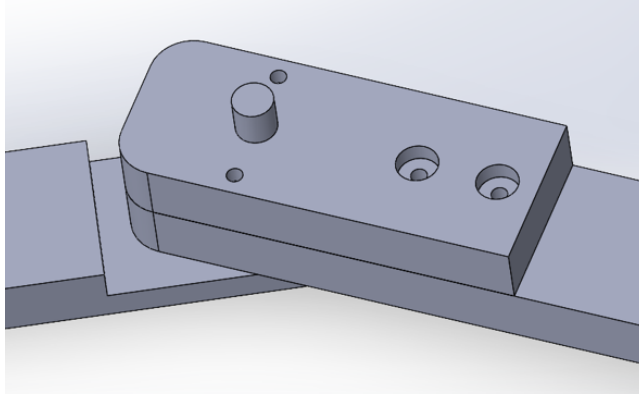


Figure 6: Encoder Mount attached on Pendulums

3.2 Bill of Materials

The primary components of the physical system are summarized in Table 1 below. The total procurement cost for the system is summarized in Table 1 below. It should be noted that several components, including the mounting brackets and base plate, were reused from a previous iteration of this project and are therefore not reflected in the total cost.

Table 1: Component Cost Summary

Item	Qty	Cost per Item	Total Cost	Vendor
Slip Ring	1	\$455.00	\$455.00	Hawk USA/MOOG
Rotary Shaft	1	\$12.16	\$12.16	McMaster-Carr
Shaft Coupler	1	\$14.99	\$14.99	Amazon
Large Steel Ball Bearing	1	\$7.72	\$7.72	McMaster-Carr
Pendulum Aluminum Bar	2	\$4.18	\$8.36	McMaster-Carr
Small Steel Ball Bearing	1	\$7.30	\$7.30	McMaster-Carr
Base Plate Aluminum Bar	1	\$15.43	\$15.43	McMaster-Carr
Slip Ring Mount Aluminum Bar	1	\$66.98	\$66.98	McMaster-Carr
Socket Head Screw	4	\$8.38	\$33.52	McMaster-Carr
Total			\$621.46	

4 System Assembly

With all custom components fully fabricated, the system was assembled and integrated into a complete working system.

4.1 Assembly Challenges

The assembly process resulted in some unexpected challenges. Upon initial fitment of the machined components, several parts did not come together as designed. Minor tolerancing discrepancies between the CAD model and the fabricated parts resulted in alignment issues in the base plate and motor mount. However, all fitment issues were resolved during assembly, with some holes having to be re-drilled and tapped in-house. The final assembled system is shown in the figures below.

4.2 Final Assembly

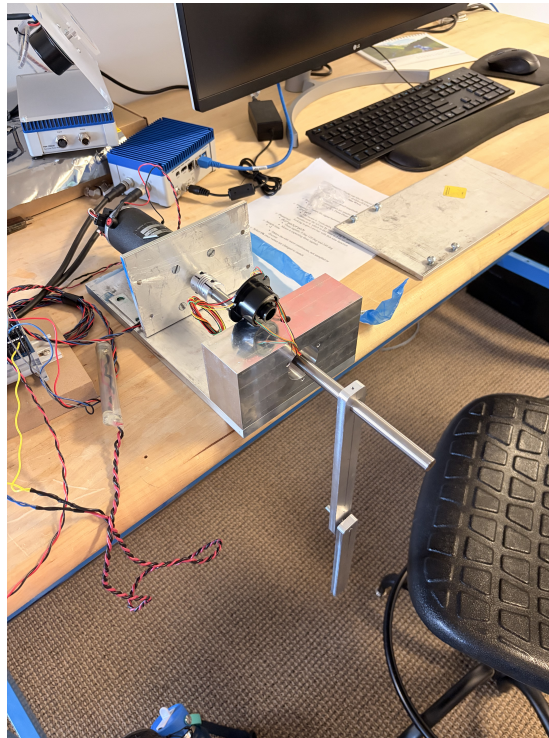


Figure 7: Completed Double Pendulum Assembly

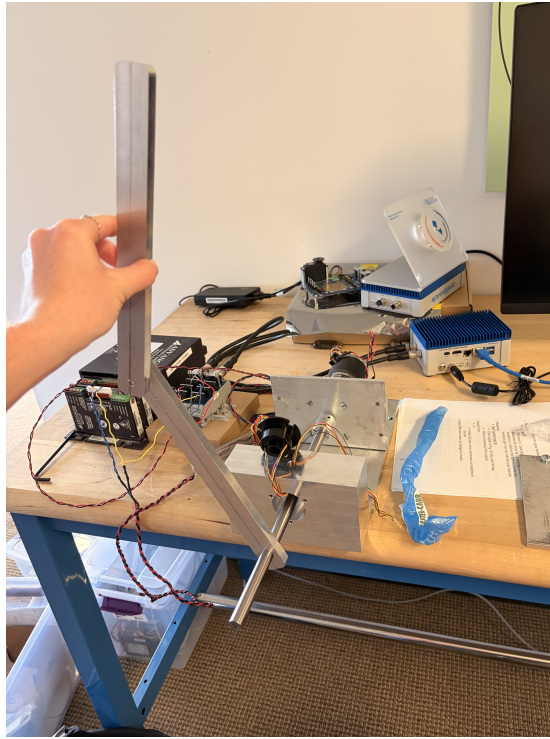


Figure 8: Double Pendulum Assembly with Pendulum Expansion



Figure 9: Encoder mount on Second Pendulum

The shaft was cut shortly after these photos were taken and was deemed operational enough for testing to begin on applying the control system.

5 SpeedGoat Configuration

Configuring the double pendulum control model onto the SpeedGoat real-time target machine required both the hardware and software to communicate correctly. The SpeedGoat Unit real-time target machine runs a real-time operating system with MATLAB Simulink through an Ethernet connection with both the host and target machine configured to internal static IP addresses. Connection was established between the two by typing `srltExplorer` in the MATLAB command window and initiating the connection.

The hardware used was the IO397 FPGA module paired with the terminal board, allowing direct access to individual I/O pins for reading the encoder signals. The Simulink model was configured as a real-time application by selecting the SpeedGoat Unit target under the apps tab in Simulink. The model was then converted to C/C++ code automatically and deployed to the target machine using the Build Application and Run on Target commands. To set up the double encoder configuration, first the brushed DC motor was controlled using these platforms. Following instructions made by a student who worked on a previous iteration of this project, the motor was successfully operated and confirmed the correct configuration of the software. The test involved arranging a PWM generation block with a three-phase sine wave input at the appropriate frequency and amplitude for the motor using the `speedgoat_IO397_TPI6020.mat` bitstream file to configure the FPGA pinout.



Figure 10: Speedgoat Real-Time Target Machine

References

- [1] A. V. Oppenheim and G. C. Verghese, *Signals, Systems and Inference*, Chapter 6: State Observers and State Feedback. MIT, 2010.

- [2] MIT OpenCourseWare, *Dynamics*, Course 16.07, Fall 2009, Lecture 20. [Online]. Available: <https://ocw.mit.edu/courses/16-07-dynamics-fall-2009/>
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